Table 4.1 Example of a timeseries data from a Milesight AM319 environmental sensor in my home.

Sensor Id	Timestamp	CO2	Humidity	PIR	PM10	PM25	Pressure	Temperature	TVOC
24e124710b423527	1749945584458	639	67	trigger	6	5	1016.6	16.4	1.2
24e124710b423527	1749946183884	636	66.5	idle	4	4	1016.6	16.4	1.2
24e124710b423527	1749946784664	649	66.5	idle	5	5	1016.6	16.5	1.2
24e124710b423527	1749947383975	646	66.5	idle	5	5	1016.8	16.5	1.2

4.1.3 Unstructured and semi-structured data

Object data encompasses the vast array of unstructured and semi-structured content that provides essential context and documentation for physical systems. This includes technical manuals, maintenance procedures, equipment photographs, inspection videos, engineering documents, and configuration files that don't fit into tabular or temporal formats but remain important for comprehensive system understanding and operational effectiveness.

Two examples of unstructured data that I will store in my home digital twin are shown in figure 4.1. I store photographs of my water meter taken by an IoT camera for validation and calibration of the machine learning model that runs to extract the meter reading from the image, and appliance manuals and other documents that will provide important context when I start to use AI to improve decision making.

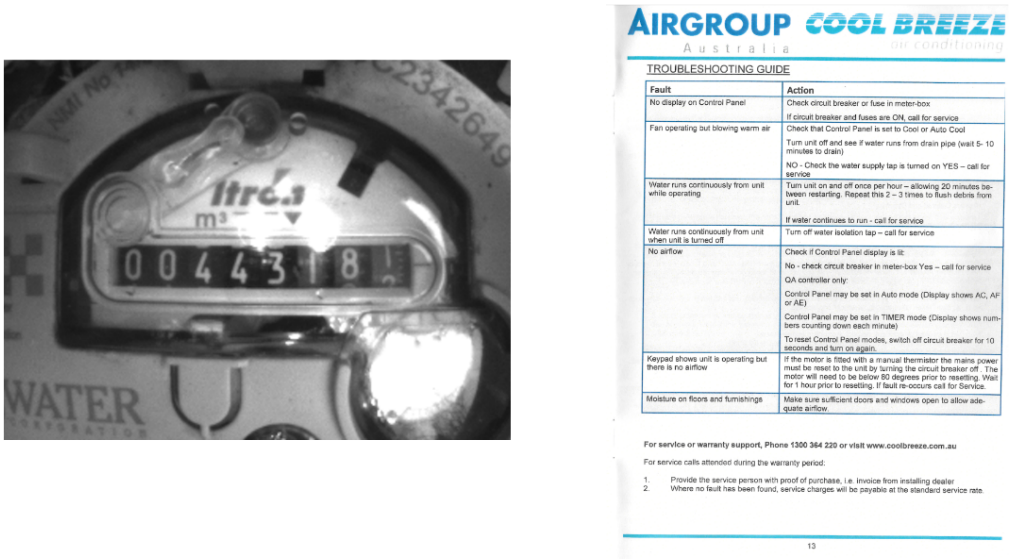


Figure 4.1 Examples of unstructured data stored in my home digital twin, with an image taken by an IoT camera shown to the left, and a device manual on the right.